



Figure 1: WSO2 architecture (Source: <https://wso2.com/platform>)

These service node components are connected to the ‘real-time event bus’ and to jClouds. The latter supports many cloud platforms like OpenStack, vCloud, CloudStack and EC2. There are also components like Artifact for distribution co-ordination, an auto scaler based on rules, a CEP (complex event processor) and cloud controllers.

If you observe closely, WSO2 is nothing but micro services based architecture that supports various services. Interaction between inner architecture and outer architecture components is made easy with messaging channels, cluster based micro services and the message bus.

WSO2 has the advantage of fostering agility and flexibility with open source. This helps in recomposing code, or making changes and additions to meet specific requirements as and when they evolve. Integration of applications is also made easy.

WSO2 helps in building solutions in the following domains:

- Financial sector (open banking)
- Regulatory compliance (GDPR or General Data Protection Regulations)

WSO2 features: These are listed below.

API management: This API manager involves three components such as API Publisher, API Gateway and API Subscriber. The WSO2 API Manager is a 100 per cent open source enterprise-class solution that supports API publishing, life cycle management, application development, access control, rate limiting and analytics in one cleanly integrated system. This allows users to design, compose and publish the APIs and, hence, API discovery is made easy.

Integration: This brings digital transformation into a single package for connecting enterprise systems and data. This feature helps in:

- Optimising business processes
- Reducing costs and potential bottlenecks
- Leveraging the technology and cost savings of the cloud
- Integrating legacy systems instead of total replacements
- Building businesses that can adapt at speed to changing conditions

IAM (identity and access management): This feature helps in securing digital business by connecting and managing multiple identities. IAM can be integrated with Azure or Salesforce based cloud solutions. IAM bridges multiple, single sign-on (SSO) protocols such as OpenID Connect, SAML 2.0 and WS-Federation to provide a unified SSO experience. It provides strong authentication and enforces multi-factor authentication with SMS/email one-time passwords (OTPs), Fast Identity Online (FIDO), MePIN, Duo Security and more.

The advantages of this feature are:

- Allows easy access from anywhere
- Connects everyone to everything
- Improves productivity
- Enhances the user experience
- Single sign-on and identity federation
- Identity based governance and administration

Analytics: You can leverage streaming analytics to gain the business intelligence you need for digital transformation. WSO2 Data Analytics Server (WSO2 DAS) is a powerful open source analytics platform that analyses data streams in real-time. It offers streaming analytics capabilities, complex event processing and machine learning to help you understand events, map their impacts, identify patterns and react within milliseconds, in real-time.

The advantages are that you can:

- Monitor and analyse with operational analytics
- Create better customer experiences
- Make informed business decisions
- Extend the solutions to analyse the past, present and future

Why WSO2 is enterprise grade: WSO2 provides many enterprise grade advantages and features that no other platform or framework does. These help in easy deployment and integration.

The advantages of WSO2 are:

- Centralised metering and monitoring with a unified logging framework
- Ability to plug in any third party health checking/monitoring framework
- Cartridge model enables bringing in even legacy apps into the cloud as service nodes
- Supports DevOps tooling
- Elastic scaling (not only HTTP based services)
- Cloud bursting (scales across multiple infra clouds simultaneously)
- Multi-zone/data centre support
- Multiple tenant isolation levels (VM, LXC, Docker)

HPE Helion Stackato, a hybrid cloud solution

Stackato (which was a Cloud Foundry and Docker based PaaS) was acquired by HPE around July 2015. HPE Helion Stackato is a cloud native platform that provides enterprises the right services, tools and control to enable developers to accelerate their innovations. HPE Helion Stackato is a polyglot Platform-as-a-Service (PaaS) and also supports SaaS. This can be used